

Harshit Arvind Barde

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Summary

AI Engineer specializing in Large Language Models, Retrieval-Augmented Generation (RAG), and AI agent orchestration. Experienced in building scalable AI systems using Python, LangChain, LangGraph, FastAPI, Apache Spark, and Airflow across AWS, Azure, and Google Cloud. Proven track record of translating complex operational data into intelligent systems that improve engineering diagnostics, volunteer coordination platforms, and telecommunications analytics.

Education

University of Michigan – Dearborn

Master of Science in Data Science

Dec 2024

Nagpur University

Bachelor of Engineering in Computer Science

Jun 2022

Experience

HCLTech (Ford Motor Company)

May 2025 – Present

AI / Machine Learning Engineer — Automotive Systems

- Reduced engineering investigation overhead by 32% by architecting Retrieval-Augmented Generation pipelines using Python, LangChain, and vector embeddings that combine VIN telemetry, recall documentation, and engineering knowledge bases.
- Accelerated root-cause identification across vehicle diagnostics by orchestrating AI agent workflows through FastAPI microservices and LLM reasoning pipelines.
- Stabilized large-scale automotive analytics processing millions of vehicle telemetry records daily by implementing distributed data pipelines using Apache Spark.
- Improved reliability of telemetry ingestion workflows by orchestrating automotive data pipelines using Apache Airflow across cloud infrastructure.

Sayaam for All

Feb 2025 – May 2025

Machine Learning Engineer

- Improved volunteer assignment efficiency by 40% by developing intelligent matching pipelines that analyze volunteer participation logs and organizational activity records using Python analytics workflows.
- Strengthened financial transparency by engineering donation logging pipelines processing structured transaction records for reporting and monitoring.
- Enabled contextual analysis of donation activity by implementing RAG-based retrieval pipelines using embeddings and vector database search.

University of Michigan – Dearborn

Oct 2023 – Dec 2024

Research Software Engineer

- Developed DeepCanvas, a CAN bus visualization and detection framework that converts raw vehicle network traffic into structured feature maps for cybersecurity research.
- Improved anomaly detection accuracy to 92% by constructing machine learning pipelines that analyze CAN message frequency patterns and network behavior.
- Processed large-scale vehicular telemetry datasets using Python and Apache Spark to support vehicle intrusion detection research.

TietoEvry India Pvt Ltd

Jan 2022 – Dec 2022

Software Engineer — Telecommunications

- Accelerated telecom fraud investigation workflows by 30% by building distributed Spark pipelines that analyze call detail records and transaction logs.
- Improved fraud detection precision to 92% by implementing feature engineering pipelines using PySpark across telecom datasets.
- Scaled telecom analytics processing across cloud infrastructure by deploying distributed data pipelines on AWS.

Project

Autonomous AI Agent Framework

- Reduced multi-step analytical processing time by 45% by designing an AI agent architecture using LangGraph and LangChain to orchestrate planner, executor, and reasoning workflows.
- Implemented Retrieval-Augmented Generation pipelines integrating document ingestion, embedding generation, and vector database retrieval.
- Developed tool-driven LLM agents capable of interacting with APIs, datasets, and external services to execute complex reasoning tasks.

Technical Skills

Languages: Python

AI/ML: Large Language Models, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, LangChain, LangGraph

Vector Databases: FAISS, Pinecone, Chroma

Data Engineering: Apache Spark, PySpark, Apache Airflow

Backend: FastAPI, Microservices

Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform